

NEGHA NELSON A

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SKILLS

Languages: Java, HTML, CSS

Databases & Frameworks: React, MongoDB, FastAPI

Developer Tools: Git, GitHub, Visual Studio Code

Core Subjects: Data Structures and Algorithms, Object-Oriented Programming, Web Development

EXPERIENCE

Software Engineer Intern

August 2024 – March 2025

Redintek Solution

Coimbatore, India

- Spearheaded backend microservices development using **FastAPI** and **PostgreSQL**, optimizing database queries and reducing API request latency.
- Collaborated with a five-member engineering team to resolve system bottlenecks and debug critical code paths, maintaining **99.8% application uptime**.

PROJECTS

Diabetic Retinopathy Detection System

November 2025 – December 2025

Python, OpenCV

- Engineered a deep learning image classification pipeline to analyze retinal scan imagery for early automated diagnosis of Diabetic Retinopathy.
- Utilized **OpenCV** to build automated image preprocessing workflows, achieving **94% classification accuracy**.

Distributed Emergency Alert and Response System

March 2024 – April 2024

React, MongoDB

- Constructed a real-time incident reporting application capable of handling **500+** concurrent user events using **React**.
- Architected secure Application Programming Interfaces (APIs) and optimized **MongoDB** schemas to reduce system data query response times by **30%**.

Bus Booking Portal

June 2023 – July 2023

HTML, CSS

- Developed a responsive UI replicating end-to-end user workflows, including real-time filter searching and seat allocation.
- Architected clean, semantic **HTML** page structures and modular **CSS** stylesheets, improving layout rendering consistency and load times.

ACHIEVEMENTS & LEADERSHIP

- **Head of Media, Events & Outreach – IIC Karunya:** Directed marketing, student engagement, and event operations for campus-wide technical initiatives at the Innovation and Incubation Cell.
- **Workshop Lead at MindKraft 2026:** Organized and conducted an AI workshop titled "Anime AI Awakening – Code the Hero Within" at Karunya's international techno-management fest, training students on Agentic AI and NVIDIA tools.
- **Technical Club Recruitment Leader:** Coordinated and executed recruitment drives for student-run technical clubs under the Division of Computer Science and Engineering in collaboration with Bytes and Clouds.

CERTIFICATIONS

- **Java Programming Fundamentals** – Infosys Springboard
- **Oracle Database Foundations Certified** – Oracle
- **Prompt Engineering for ChatGPT** – Great Learning
- **Artificial Intelligence: Types of AI** – Infosys Springboard
- **Networking Essentials** – Cisco Networking Academy

EDUCATION

Karunya Institute of Technology and Sciences

Expected May 2027

Bachelor of Technology in Computer Science and Engineering – CGPA: 8.14

Coimbatore, India